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Welcome for the training on Techno functional training program

Agenda:
1. Introduction
2. LET portal
3. Day schedule (9 30am to 630pm)

• 11 30 (20 minutes)
• Lunch 1 30 (60 minute)
• 4 (20 minute)
4. Lab guidance
5. Courseware
6. TOC

9664670840

• https://linuxlab.koenig-solutions.com
• Pa$$w0rd
==========
Apllication development:
Monolithic:
Microservice:

components:
1)frontend ==>dotnet==>EFcore

==========
SQL server==>DB server

SSMS ==>Db client

CRUD operation

blameless postmortum

code===>build===>test==>deploy===>maintain
==========
Account required for training:
---
github
docker hub
redhat
==========
Database
UI

DTO===>ORM

==========

Day 2: 20/10/20

https://github.com/Mudassir-dev0ps/dotnetSampleApp

MVC
==========
```
using Microsoft.EntityFrameworkCore;
using OrderManagement.Models;

namespace OrderManagement.Data
{
 public class ApplicationDbContext : DbContext
 {
 public ApplicationDbContext(
 DbContextOptions<ApplicationDbContext> options)
 : base(options)
 {
 }

 public DbSet<Customer> Customers { get; set; }

 public DbSet<Order> Orders { get; set; }
 }
}
```

=
dll===> format===>published application
==========
```
var builder = WebApplication.CreateBuilder(args);

var app = builder.Build();

app.MapGet("/", () =>
"Hello! My .NET application is running.");

app.MapGet("/about", () => new
{
 Application = "BuildPublishDemo",
 Technology = "ASP.NET Core",
 Status = "Running"
});

app.MapGet("/health", () => new
{
 Status = "Healthy",
 Time = DateTime.UtcNow
```

});

app.Run();

==========

Day 3: 21/10/23

Day 6 – Windows Server Fundamentals
•
Windows Server architecture
•
Windows Server editions
•
System configuration
•
Windows Server roles
•
Server Manager
•
Users and groups
•
Local vs domain accounts
•
Active Directory overview
•
Group concepts
•
User management
•
Hands-on user/group administration

==========
OS
dependencies
libraries
App code
==========

Identity management ===>Authentication==========>IAM
Access managennt====>Authorization

AD==========>Windows
LDAP==========>Linux
Entra ID====>Azure
IAM===>AWS
==========
The participants will act as junior Windows Server administrators for a fictional company. They will configure a server, install roles, create users and groups, deploy Active Directory, and troubleshoot access and account-management issues.

1 role==>UI
1 role==powershell
---
local user
2local group

AD
local user
2local graoup

==========

Day 3 22/10/26

Windows Filesystem & Permissions

- Windows filesystem
- Drives and partitions
- NTFS
- File and folder permissions
- NTFS permissions
- Share permissions
- Permission inheritance
- Ownership
- Administrative shares
- Disk space troubleshooting

==========
3GB====10x3==1000
3GiB===2x10=3072
==========
Folder1==========>f1,f2===>content ==========>UI
Folder2==========>f3,f4========>content==========>Powershell

- Access: Folder FIles
  Read
  Write
  Execute

ACL====> access control list
ACE

icacls (powershell command for ACL access)
https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/icacls#:~:text=A%20sequence%20of%20simple%20rights,%2D%20Read%20and%20execute%20access

#icacls C:\NTFSDemo\Finance /grant "FinanceLab:(OI)(CI)M"

Learning principle
1)simple to complex
2)real to imaginary
3)part to whole

==========
Ownership:

RDP

run -> sysdm.cpl ->remote-> allow remote connection to this computer

RDP- Enable-NetFirewallRule -DisplayGroup "Remote Desktop" | Select-Object DisplayName, Enabled, Profile

Add user to Remote Desktop User group in win22 VM

==========
Network file sharing

SMB

oct19jr-tfaodp
Pa$$w0rdR# RDP- Enable-NetFirewallRule -DisplayGroup "Remote Desktop" | Select-Object DisplayName, Enabled, Profile

---

Administrative shares:

C$ Access the C: drive remotely

D$ Access the D: drive remotely, if available

ADMIN$ Access the Windows installation directory, usually C:\Windows
==========

Day5 23/10/26

Windows Processes & Services

Processes
Threads Services
Service dependencies
Windows Services console
Task Manager
Resource Monitor
CPU troubleshooting
Memory troubleshooting
Service startup failures
Service recovery
==========
Process vs Program

multiprogramming vs multiprocessing

windows==>run===>dxdiag

Process & Parent Process ->
Get-CimInstance Win32_Process | Select-Object Name, ProcessId, @{Name="ParentName"; Expression={(Get-CimInstance Win32_Process -Filter "ProcessId = $($_.ParentProcessId)").Name}}, ParentProcessId | Format-Table -AutoSize

tasklist /svc

Types of processes:
1)foreground: 80% resource
2)background: 20 % resource
==========
Process vs. Daemons
Service==========>daemon
Restart/Reload
---
100 request
True positive /False positive True False
Triage

Multithreading demonstration

```
------------------
import threading, time

def task(n):
 print(f"Task {n} started")
 time.sleep(2) # simulates waiting (API/DB call)
 print(f"Task {n} done")

start = time.time()
threads = [threading.Thread(target=task, args=(i,)) for i in range(4)]
for t in threads: t.start()
for t in threads: t.join()
print(f"Total: {time.time() - start:.1f}s") # ~2s, not 8s
```

==========

Multiprocessing .

```
import multiprocessing, time

def burn_cpu(n):
 total = 0
 for i in range(30_000_000):
 total += i * i
 print(f"Process {n} done")

if __name__ == "__main__": # required on Windows
 start = time.time()
 procs = [multiprocessing.Process(target=burn_cpu, args=(i,)) for i in range(4)]
 for p in procs: p.start()
 for p in procs: p.join()
 print(f"Total: {time.time() - start:.1f}s")
```

==========

Python Releases for Windows | Python.org

```
1 - $url = "https://www.python.org/ftp/python/3.14.2/python-3.14.2-amd64.exe">> Invoke-WebRequest -Uri $url -OutFile "$env:TEMP\python-installer.exe"

2 - $pythonDir = "$env:LOCALAPPDATA\Programs\Python\Python314"
$scriptsDir = "$pythonDir\Scripts"

$userPath = [Environment]::GetEnvironmentVariable("Path", "User")
$entries = @($userPath -split ';' | Where-Object { $_ })

foreach ($dir in @($pythonDir, $scriptsDir)) {
 if ($entries -notcontains $dir) {
 $entries += $dir
 }
}

[Environment]::SetEnvironmentVariable(
```

"Path",
 ($entries -join ';'),
"User"
)

$env:Path = "$pythonDir;$scriptsDir;$env:Path"
==========
event=====>incident

logs===>low level
event========>high level

Logs/event
severity/priority==========>importance
source==========>who
==========

- **Command Purpose**

Get-Process Lists running processes
Get-Service Lists services
Get-CimInstance Win32_Process Retrieves process details
Get-CimInstance Win32_Service Retrieves service configuration and status
Get-Counter Reads performance counters
Get-WinEvent Reads Windows event logs
Start-Service Starts a service
Stop-Service Stops a service
Restart-Service Restarts a service
Set-Service Changes supported service settings
Get-Process -Id Inspects a process by PID
==========

Day 6: 24/10/26

Windows Logs, IIS & Firewall

Event Viewer
Windows event logs
Application logs
System logs
Security logs
IIS fundamentals
IIS architecture
Websites
Application pools
IIS bindings
IIS logs
Windows Firewall
Inbound/outbound rules
==========

```
<configuration>
<system.webServer>
<handlers>
<add name="httpPlatformHandler" path="*" verb="*"
 modules="httpPlatformHandler" resourceType="Unspecified"/>
</handlers>

<httpPlatform processPath="C:\inetpub\myproject\venv\Scripts\python.exe"
 arguments="-m waitress --port=%HTTP_PLATFORM_PORT% myproject.wsgi:application"
 stdoutLogEnabled="true"
 stdoutLogFile="C:\inetpub\myproject\logs\python.log"
 startupTimeLimit="60"
 requestTimeout="00:02:00">
<environmentVariables>
<environmentVariable name="DJANGO_SETTINGS_MODULE" value="myproject.settings"/>
</environmentVariables>
</httpPlatform>
</system.webServer>
</configuration>
```
==========

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https://github.com/Mudassir-dev0ps/dotnetSampleApp

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Day 7: 26/10/26

Agenda:
PowerShell & Windows Troubleshooting
PowerShell fundamentals
Cmdlet
Variables
Pipelines
Objects
Services through PowerShell
Processes through PowerShell
Filesystem automation
Network troubleshooting commands
Windows performance troubleshooting
Production troubleshooting scenarios
==========
Cmdvs Powershell

#To get all attribute of the command
Get-Process | Get-Member -MemberType Property

==========
Automate following using powershell script:
Services through PowerShell
Processes through PowerShell
Filesystem automation
Network troubleshooting commands
Windows performance troubleshooting
==========
Security:

System hardening

System security:
DAC: user management and permission
MAC: SELinux

Defender:
inbond
outbond
==========
==========